

Davide Beltrame

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EDUCATION

Bocconi University

MSc in Artificial Intelligence (LM-18 Computer Science)

Milan, Italy

Sep 2024 – Oct 2026

- **Achievements:** Best-in-class graded exam projects in NLP and Computer Vision; selected for the advanced course track and admitted to two PhD-level courses on statistical and applied ML.
- **Societies:** *Bocconi Students for Machine Learning (BSML)* Advisory Board & Researcher; *Bocconi Open Source Society (BOSS)* Co-founder; *Consensus Rivista* STEM Editor-in-Chief.

LUISS Guido Carli University

BSc in Management and Computer Science – Grade: 110/110 cum laude

Rome, Italy

Sep 2021 – Jul 2024

- **Merit-based Scholarship:** Full tuition coverage (ranked 1st among applicants); maintained top-class ranking with 8 “laude” grades (31/30).
- **Thesis:** *ML for Eco-climatic Strategy: Predicting Air Transport Emissions*. Supervisor: Prof. Giuseppe F. Italiano.
- **Societies:** Starting Finance Club Guido Carli President (2023-2024); Junior Enterprise LUISS (JELU) Auditor (2022-2024).

RESEARCH EXPERIENCE

MSc Thesis Researcher

Bocconi Machine Learning & Statistical Physics Lab (Supervisor: Prof. Luca Biggio)

Milan, Italy

Sep 2025 – Present

- Extending the lab’s recent work (Garnier-Brun J., Biggio L., Saglietti L., Mézard M. *Early-stopping Too Late?*, presented at MemFM@ICML 2025) on generalization in diffusion models.
- Focusing on the inductive bias of different architectures; designing experiments to isolate how structural priors influence the transition phase between generalization and memorization in hierarchical data models.

Student Researcher

MilaNLP Lab (Supervisors: Prof. Dirk Hovy, Dr. Paul Röttger)

Milan, Italy

Sep 2025 – Present

- Developing a zero-shot length predictor for masked diffusion models (MDMs) to eliminate the “padding tax” (computational waste from fixed-length context windows) based only on the internal representations of the model.
- Achieved 60–87% FLOPS reduction across benchmarks (GSM8K, HumanEval, LongFormQA).

Board Member and Researcher

Bocconi Students for Machine Learning (BSML)

Milan, Italy

Apr 2025 – Present

- First-authored a study on interpretability in medical imaging (“Do AI and Human Experts See Pneumonia the Same Way?”), comparing saliency maps of CNNs with human radiologist attention (collected 300+ annotations from 15+ experts).
- Co-organized events on campus (HPC tutorial, meetups with industry contacts) open to the university community.

WORK EXPERIENCE

AI Engineering Intern

CommerceClarity

Milan, Italy

May 2025 – Sep 2025

- Engineered an end-to-end Python pipeline for automated product image compliance, reducing the manual rejection rate from 5% to 0%, unblocking the listing funnel and eliminating human-in-the-loop review costs.
- Designed company-wide analytics dashboards (Laravel/REST/SQL) to expose operational bottlenecks and real-time ingestion metrics, adopted for weekly management reviews.

Tutor

Italian Informatics Olympics (OII) Training at LUISS

Rome, Italy

Mar 2022 – May 2023

- Architected the full technical pipeline for competitive programming contests: designed 10+ original algorithmic problems, implemented robust input generators, and wrote reference solvers in Python and C++.
- Prepared lectures on computational complexity, dynamic programming and graph theory.

SKILLS & ADDITIONAL INFORMATION

Coding: Python (PyTorch, TensorFlow, Pandas, NumPy), C++, C, SQL, MATLAB, L^AT_EX.

Tools: Linux, Git, Docker, SLURM (HPC), Weights & Biases, FastAPI.

Languages: Italian (native), English (fluent C2 – IELTS 8.5), French (intermediate B2), Spanish (basic).

Volunteering: Foundation Banco Alimentare ETS (2019), APS “Il Centro” (2018-2019)

Awards: Top 1% (District Level): National Mathematics, Physics, and Informatics Olympics.

Certifications: GRE General Test (2022): Quant 169/170, Verbal 163/170 (92nd percentile overall).